

File

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 TERM 3

2013

Mathematics

General Instructions

- Working time – 75 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in Questions 6 to 13
- Start each question on a new page

Total marks – 70 marks

Section 1 – 5 marks

Attempt questions 1 -5

Allow about 7 minutes for this section

Section 2 – 65 marks

Attempt questions 6 to 13

Allow about 63 minutes for this section

Name: _____

Teacher: _____

Section I (5 marks)

Attempt questions 1 to 5.

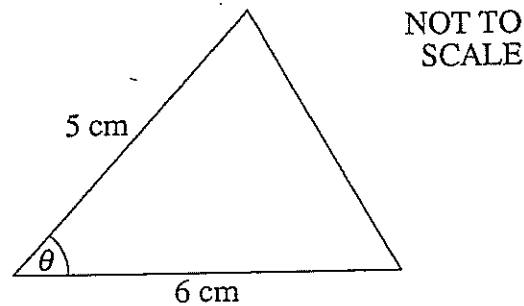
Allow approximately 7 minutes for this section.

Use the multiple choice answer sheet provided in your answer books.

Select the alternative A, B, C or D that best answers the question.

Fill in the response oval completely.

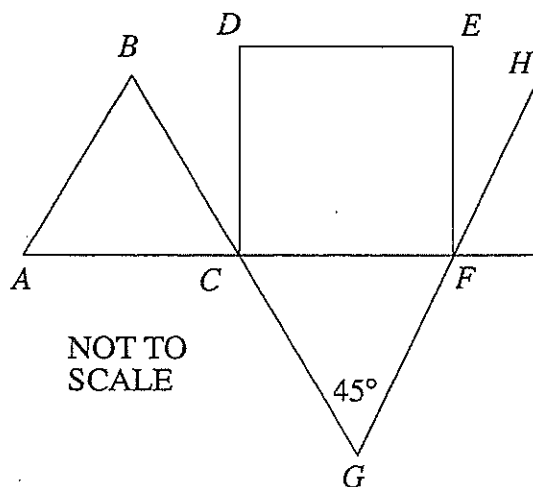
Question 1



Find the value of $\sin \theta$ if the area of this triangle is 10 cm^2 .

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

Question 2



ABC is an equilateral triangle.

$CDEF$ is a square.

BCG , ACF , and GFH are straight lines.

$\angle CGF = 45^\circ$.

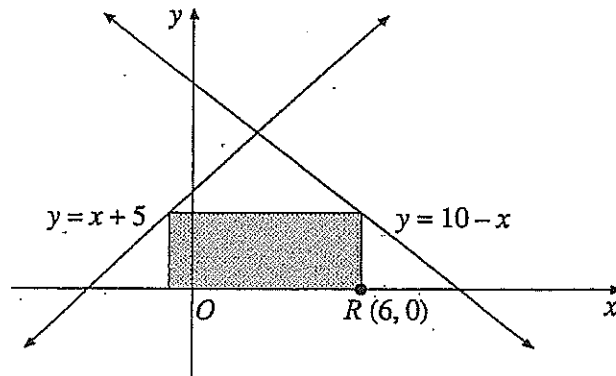
The size of $\angle EFH$ is

- (A) 15°
(B) $22\frac{1}{2}^\circ$
(C) 30°
(D) 45°

Question 3

R is the point with coordinates $(6, 0)$.

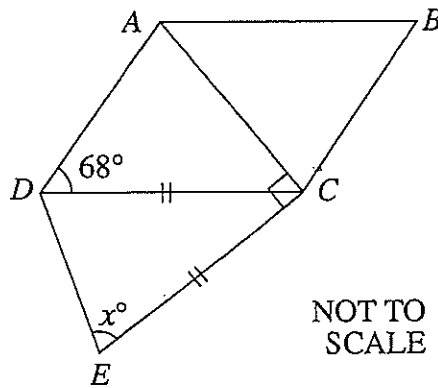
NOT TO SCALE



The area of the shaded rectangle in square units is

- (A) 20 (B) 24 (C) 28 (D) 60

Question 4



NOT TO SCALE

$ABCD$ is a rhombus.

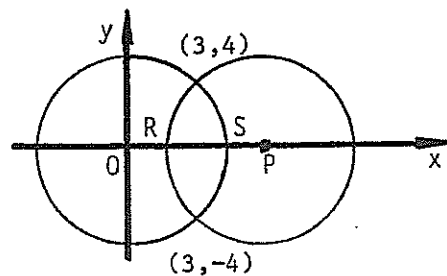
$DC = EC$.

$\angle ADC = 68^\circ$ and $\angle ACE = 90^\circ$.

The value of x is

- (A) 22
(B) 68
(C) 73
(D) 79

Question 5



O and P are centres of equal circles which intersect at $(3, 4)$ and $(3, -4)$. The circles cut the x axis as shown at R and S .

$RS = ?$

A. 1

B. 2

C. 4

D. 5

Section II - Attempt questions 6 – 13.

All answers and working to be done in your answer book.

Question 6 (8marks)

- a) Simplify giving the answer to 2 significant figures

2

$$\frac{4.7 + \sqrt{3.2}}{1 + 4.15^2}$$

- b) Solve $-3 \leq 1 - 2x < 5$

2

- c) If $(2 + \sqrt{3})^2 = a + \sqrt{b}$ Find the value of a and b

2

- d) Solve $x^2 \geq (x + 1)(x + 5)$

2

Question 7 (8 marks) (Start a new page)

- a) Factorise fully

i. $8mn^2 + 6m^2n + 10mn$

1

ii. $5x^2 - 33x + 18$

1

iii. $125 - 27x^3$

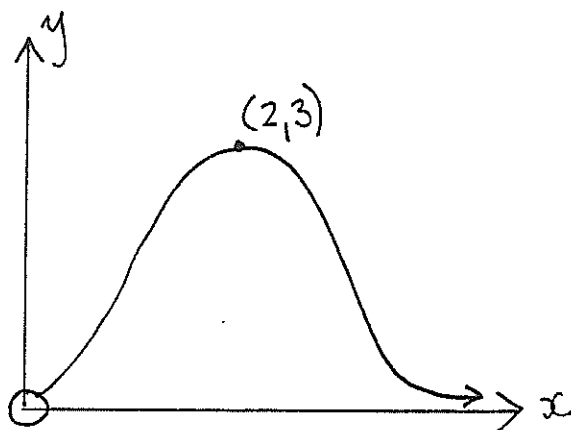
2

- b) If $\sin \theta = \frac{4}{7}$ and $\tan \theta < 0$ find the exact value of $\cos \theta$

1

c)

3



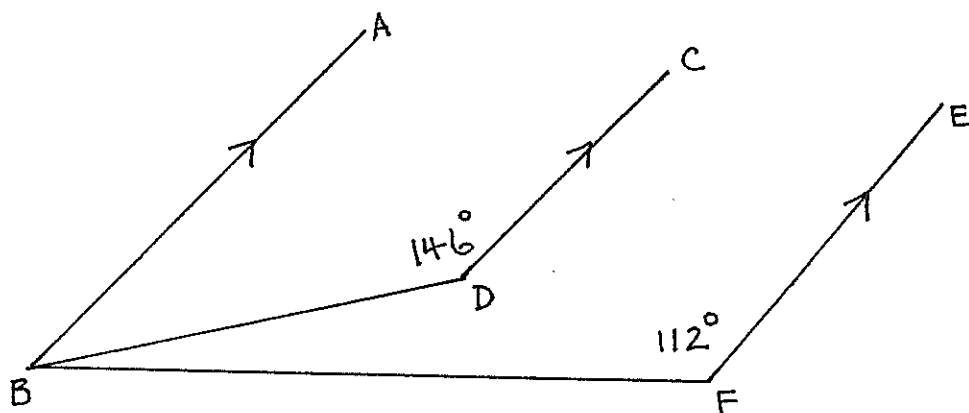
Consider the graph of $y = f(x)$

- i. State its domain
- ii. State its range
- iii. Write down the equation (s) of any asymptotes

Question 8 (9 marks) (Start a new page)

a)

3



In the diagram above $AB \parallel CD \parallel EF$, $\angle BDC = 146^\circ$ and $\angle EFB = 112^\circ$. Show, giving reasons, that BD bisects $\angle ABF$.

b) Write $\frac{1}{x+1} - \frac{1}{x-1}$ as a single fraction

2

c) Solve for x

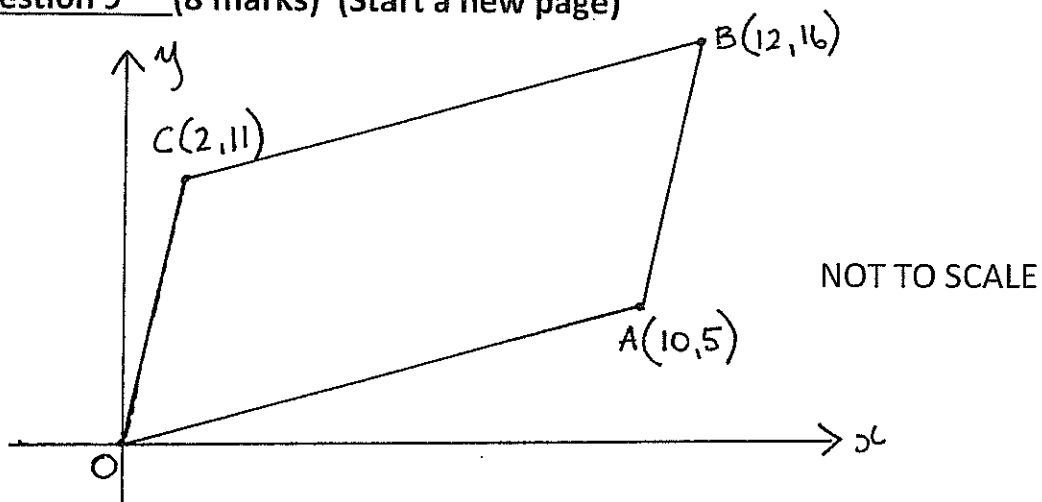
i. $\frac{5}{8}(x+4) = 4x - \frac{1}{2}$

2

ii. $x + \frac{2}{x} = 3$

2

Question 9 (8 marks) (Start a new page)



In the diagram above A, B and C are the points (10,5), (12,16) and (2,11) respectively

Copy the diagram into your answer sheets

a) Find the distance AC

1

b) Find the midpoint of AC

1

c) Show the $OB \perp AC$

2

d) Find the midpoint of OB and explain why OABC is a rhombus

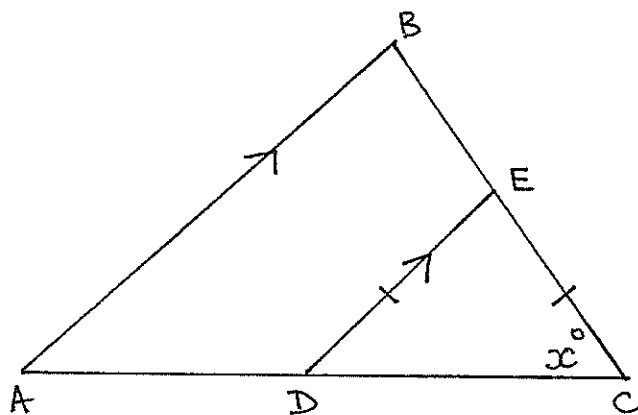
2

e) Hence find the area of OABC

2

Question 10 (8 marks) (Start a new page)

a)



In the diagram above $AB \parallel DE$ and $DE = EC$. Let $\angle ECD = x^\circ$

- i. Copy the diagram above onto your answer sheets
 - ii. Prove $AB = BC$ (reasons required) 3
- b)
- i. Sketch $y = |x - 1|$ and $y = |x|$ on the same axes
(Label each graph carefully) 2
 - ii. Hence solve $|x - 1| = |x|$ 1
- c) Solve $|x + 3| = 2x - 1$ 2

Question 11 (8 marks) (Start a new page)

- a) Find the acute angle θ , correct to the nearest minute,
given that $\tan \theta = 1 \frac{1}{3}$ 2
- b) Simplify $\frac{\cos (90 - \theta)}{\cos (180 - \theta)}$ 2
- c) Solve $2\sin \theta - 1 = 0$ for $0 \leq \theta \leq 360^\circ$ 2
- d) Simplify $\sec^2 \theta - \tan^2 \theta$ 2

Question 12 (8 marks) (Start a new page)

- a) Prove the identity 3

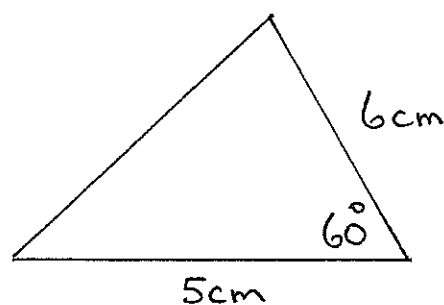
$$\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta} = 2\sec^2 \theta$$

- b) Solve the following trigonometric equations in the domain $0^\circ \leq \theta \leq 360^\circ$
- i. $3\cos^2 \theta + 5\cos \theta = 0$ 3
- ii. $\sqrt{3} \sin \theta - \cos \theta = 0$ 2

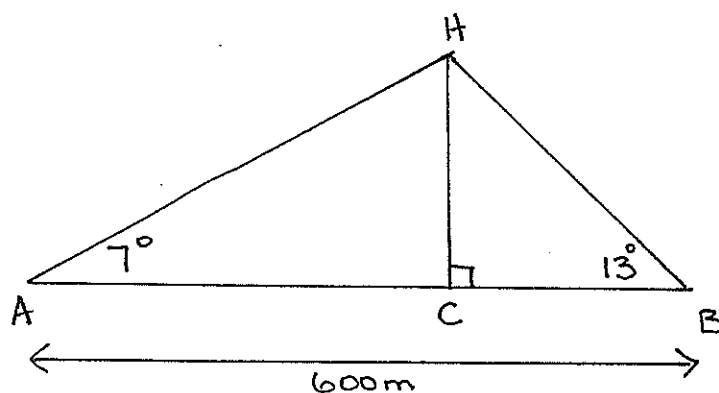
Question 13 (8marks) (Start a new page)

a) Find the exact area of the triangle below.

1



b) In the diagram below, a helicopter H is hovering above a straight, horizontal road AB of length 600 metres. The angles of elevation of H from A and B are 7° and 13° respectively. The point C lies on the road directly below H.



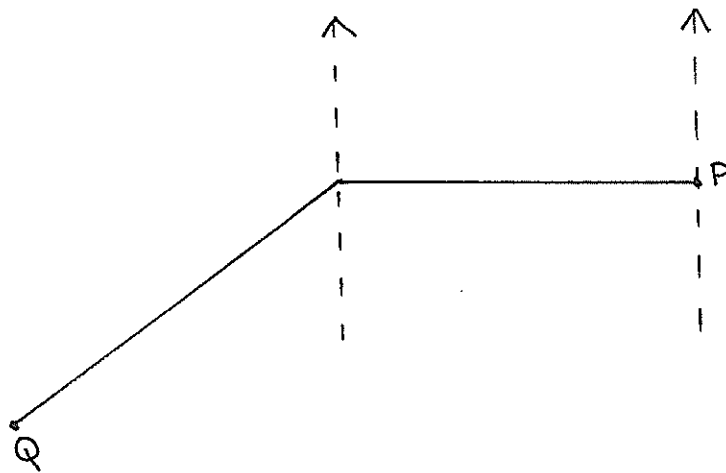
i. Use the sine rule to show that $HB = \frac{600 \sin 7^\circ}{\sin 160^\circ}$

1

ii. Hence, find the height CH of the helicopter above the road, correct to the nearest metre.

1

- c) A ship sails 24 km west of port P and then 17 km on a bearing of $S 42^{\circ}17' W$ to port Q.



- i. Copy the diagram above onto your answer sheets and show the information on it.
- ii. Calculate the distance of Q from P to one decimal place. 2
- iii. Calculate the TRUE bearing of the ship from P (to the nearest degree) 3

MATHEMATICS

SOLUTIONS YR 11 TERM 3 (WEEK 2) 2 UNIT

Question 1 B

Question 2 A

Question 3 C

Question 4 C

Question 5 C

Question 6

a) 0.36 (2 sig. fig)

b) $-3 < 1 - 2x < 5$

$-3 < -2x$ $1 - 2x < 5$

$2x \leq 4$ $-4 < 2x$

$x \leq 2$ $-2 < x$

$-2 < x \leq 2$

c) $(2 + \sqrt{5})^2 = a + \sqrt{b}$

$4 + 4\sqrt{5} + 5 = a + \sqrt{b}$

$9 + 4\sqrt{5} = a + \sqrt{b}$

$a = 9$ $b = 48$

$x^2 > (x+1)(x+5)$

$x^2 > x^2 + 6x + 5$

$-6x > 5$

$x \leq -5/6$

Question 7

a) i) $2mn(4n + 3m + 5)$

ii) $(5x-3)(x-6)$

iii) $(5-3x)(25+15x+9x^2)$

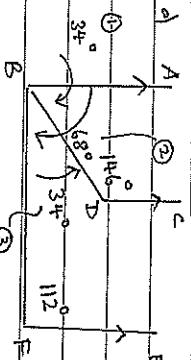
b) $\frac{1}{\sqrt{33}}$ $\therefore \cos \theta = \frac{\sqrt{33}}{7}$

c) i) D. $x > 0$

ii) R. $0 < x \leq 3$

iii) $y = 0$ (asymptote)

Question 8



$\angle ABD = 34^\circ$ (corresponding angles)

$AB \parallel CD$

$\angle BFE = 68^\circ$ ($AC \parallel EF$, corresponding angles)

$\therefore \angle BFE = 34^\circ$

$\therefore BD$ bisects $\angle ABC$

b) $\frac{1}{x+1} - \frac{1}{x-1} = \frac{x-1-(x+1)}{(x+1)(x-1)}$

$= \frac{-2}{x^2-1}$

c) i) $\frac{5}{8}(x+4) = 4x - \frac{1}{2}$

$5(x+4) = 32x - 4$

$5x + 20 = 32x - 4$

$24 = 27x$

$x = \frac{8}{9}$

ii) $x + \frac{2}{x} = 3$

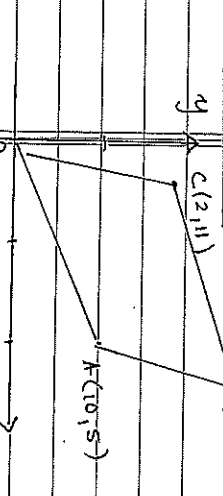
$x^2 + 2 = 3x$

$x^2 - 3x + 2 = 0$

$(x-2)(x-1) = 0$

$\therefore x = 2, x = 1$

Question 9



a) $AC = \sqrt{(10-2)^2 + (5-11)^2}$

$= \sqrt{64 + 36}$

$= \sqrt{100}$

$AC = 10$ units

b) Midpt AC (6, 8)

c) $m = \frac{16}{-8} = -2$

$m = \frac{6}{-8} = -\frac{3}{4}$

$\frac{4}{3} \times -\frac{3}{4} = -1 \therefore OB \perp AC$

d) Midpt OB (6, 8)

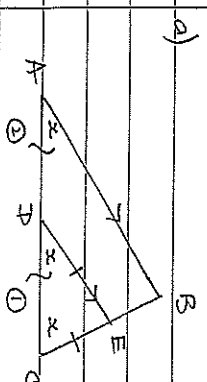
both diagonals bisected and meet at $AO \therefore OABOC \rightarrow$ rhombus

e) $OB = \sqrt{11^2 + 16^2}$

$OB = 20$ units

$\therefore \text{Area } OABC = \frac{20 \times 10}{2} = 100 \text{ unit}^2$

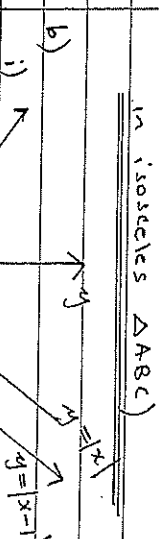
Question 10



$\angle EDC = \angle ECD$ (opposite equal sides in isosceles $\triangle DEC$)

$\angle BAD = \angle EDC$ (corresponding angles $AB \parallel DE$)

$\therefore AB = BC$ (opposite equal angles in isosceles $\triangle ABC$)



i) $y = x - 1$

ii) $x = \frac{1}{2}$

c) $|x+3| = 2x-1$

$x+3 = 2x-1$ $x+3 = -(2x-1)$

$A = x$ $x+3 = -2x+1$

$3x = -2$

$x = -\frac{2}{3}$

check solutions.

$|4+3| = 8-1$ $|-2/3+3| = 2(-2/3)-1$

$7 = 7$ $2\frac{1}{3} \neq -2\frac{2}{3}$

$\therefore$ true $x = -\frac{2}{3}$ only solution

Question 11

a) $\tan \theta = \frac{1}{3}$ $\theta = 53^\circ 8'$
(nearest min required)

b) $\frac{\cos(90-\theta)}{\cos(180-\theta)} = \frac{\sin \theta}{-\cos \theta}$
 $= -\tan \theta$

c) $2\sin \theta - 1 = 0$
 $\sin \theta = \frac{1}{2}$ $\checkmark$ S/A $\checkmark$
acute $\theta = 30^\circ$
 $\therefore \theta = 150^\circ, 30^\circ$

d) $\frac{\sec^2 \theta - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1}{1 + \tan^2 \theta}$

Question 12

a) $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta} = 2\sec^2 \theta$

LHS = $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta}$
 $= \frac{(1-\sin \theta) + (1+\sin \theta)}{1-\sin^2 \theta}$
 $= \frac{2}{\cos^2 \theta}$
 $= 2\sec^2 \theta$
 $= \text{RHS}$

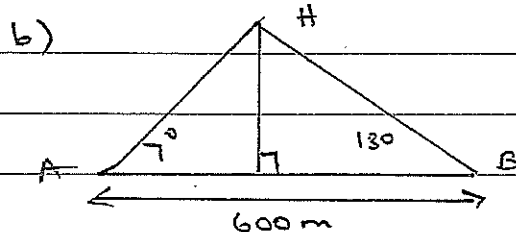
b) i) $3\cos^2 \theta + 5\cos \theta = 0$
 $\cos \theta (3\cos \theta + 5) = 0$
 $\cos \theta = 0$ $\cos \theta = -5/3$
 $\theta = 90^\circ, 270^\circ$ no solutions

ii) $\sqrt{3}\sin \theta - \cos \theta = 0$
 $\sqrt{3}\sin \theta = \cos \theta$ $\checkmark$ A $\checkmark$ acute $\theta = 30^\circ$
 $\frac{\sqrt{3}\sin \theta}{\cos \theta} = 1$ $\checkmark$ $\therefore \theta = 30^\circ, 210^\circ$
 $\therefore \tan \theta = \frac{1}{\sqrt{3}}$

Question 13

a) $A = \frac{1}{2} \cdot 6 \cdot 5 \cdot \frac{\sqrt{3}}{2}$

$A = \frac{15\sqrt{3}}{2} \text{ cm}^2$

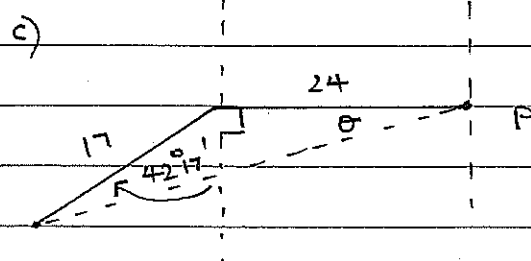


i) $\frac{HB}{\sin 7^\circ} = \frac{600}{\sin 160^\circ}$

$HB = \frac{600 \sin 7^\circ}{\sin 160^\circ}$

ii) $\sin 13^\circ = \frac{CH}{HB}$
HB $\leftarrow$ from above

$\therefore CH = HB \cdot \sin 13^\circ$
 $= 48\text{m}$



i) $QP^2 = 17^2 + 24^2 - 2 \times 17 \times 24 \times \cos(132^\circ 17')$
 $\therefore QP = 37.6\text{ km (1 dec pl)}$

ii) $\frac{\sin \theta}{17} = \frac{\sin 132^\circ 17'}{QP}$

$\therefore \theta = 20^\circ$ (to nearest deg)

true bearing ship from P = 250° T